D06 · Logistics Technology & Innovation

# 05 · AI & machine learning applications in logistics

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

## 1 WHAT IT IS

AI and machine learning in logistics apply prediction and optimization to operational data — late-shipment prediction, dynamic ETAs, freight invoice anomaly detection, capacity forecasting — to support the people who make logistics decisions.

## 2 WHY IT MATTERS

The limiting factor is rarely the algorithm; **it is the quality and accessibility of historical data, and whether anyone acts on the output**. Models trained on the past struggle with unprecedented disruptions and can amplify historical bias. The successful pattern is narrow, measurable use cases with short feedback loops, embedded in existing workflows.

## 3 HOW IT IS DONE

1. **Choose a use case with a clear decision attached.** Example: which shipments to reassign today because they are likely to be late.
2. **Check data readiness first.** History, granularity, labels and quality; fix the pipeline before the model.
3. **Pilot with a control group.** Measure the operational result of acting on predictions, not only model accuracy.
4. **Design for human adoption.** Explain recommendations, capture planner feedback and retrain on it.

### IN PRACTICE

**Begin with use cases that have short feedback loops and clear ROI**

Late-shipment prediction, invoice anomaly detection and dynamic ETA improvement are reliable starting points.

**Treat data quality as the primary investment**

No algorithm overcomes systematically incomplete or incorrect historical data.

**Design for human + AI collaboration**

Planners who understand why a recommendation was made are far more likely to act on it and to provide the feedback that improves the model.

### ILLUSTRATIVE CASE

*Illustrative case built from typical industry values — not data from a specific company.*

**Illustrative case — Distribution company implementing carrier OTIF prediction**

| Application | Model Performance | Operational Impact |
|---|---|---|
| OTIF prediction by carrier + lane + day | AUC 0.84 · 78% accuracy on delay prediction 48 h ahead | Enabled proactive reassignment |
| Proactive reassignment of critical shipments | — | +4.2 pp OTIF on high-risk days |
| Automatic alerts to customer service | — | 68% of delays communicated before customer contact |

**Result:** Customer complaints related to late delivery fell 28% in six months. The majority of the gain came from acting on predictions rather than from the predictions alone.

### HOW IT IS MEASURED

**AI Decision Support Rate %**

(Decisions of a given type that used AI input / Total decisions of that type) × 100

*Benchmark: >60% indicates meaningful adoption for the targeted decision type*

**Value Realized from AI Use Cases**

Quantified service or cost improvement attributable to AI-supported decisions

*Benchmark: Track per use case; avoid vanity metrics such as "number of models deployed"*

**WHAT YOU WOULD USE**

**AI in Logistics**

**project44 AI Insights / FourKites Intelligent Visibility** — Strong predictive ETA and risk capabilities built on large visibility data sets.

**Blue Yonder Luminate / Kinaxis / custom ML on cloud platforms** — Broader planning and optimization AI capabilities for companies ready to invest in data foundations.

## 4 SO WHAT

**Start narrow and measure decisions, not models.** Pick use cases with short feedback loops, fix the data first, pilot with a control group and design for planner trust.

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